

ETCH-Z500

Plasma Etch Tool SECS/GEM Interface Specification

NanoDyne Systems
ND-ETCHZ500-GEM-ICD-001 Rev A

Equipment Model	ETCH-Z500
Manufacturer	NanoDyne Systems
Interface	SEMI E5 SECS-II and SEMI E30 GEM
Host Link	HSMS-SS active/passive TCP/IP
Release Date	2026-06-07

1. Interface Overview

The ETCH-Z500 is a single-wafer plasma etch platform for dielectric and silicon etch processes. This document defines the GEM host interface used for equipment monitoring, event collection, remote operation, alarm handling, and process result reporting.

Item	Value
Equipment ID	ETCH-Z500-01
Default Device ID	0
T3 Timeout	45 sec
T5 Timeout	10 sec
T6 Timeout	5 sec
T7 Timeout	10 sec
T8 Timeout	5 sec
Control States	OFFLINE, LOCAL, REMOTE
Supported Streams	S1, S2, S5, S6, S7, S10

2. Host Communications

The tool supports HSMS single-session operation. The host may establish communications with S1F13/S1F14, request online state, subscribe reports to collection events, download recipes, and issue remote commands when the equipment is in REMOTE control.

3. Status Variables

Status variables are read using S1F3 and reported through linked event reports.

SVID	Name	Type	Unit	Description
1001	SystemStatus	STRING	-	IDLE, PROCESSING, FAULT, MAINTENANCE
1002	ControlState	U1	-	GEM control state 1-5
1003	ChamberPressure	FLOAT	mTorr	Process chamber pressure
1004	ChamberTemperature	FLOAT	deg C	Chamber wall temperature
1005	RFPower_Forward	FLOAT	W	Forward RF power to plasma
1006	RFPower_Reflected	FLOAT	W	Reflected RF power
1007	BiasVoltage	FLOAT	V	DC bias voltage on chuck
1008	GasFlow_CF4	FLOAT	sccm	CF4 etch gas flow rate
1009	GasFlow_O2	FLOAT	sccm	O2 additive gas flow rate
1010	GasFlow_Ar	FLOAT	sccm	Argon carrier gas flow rate
1011	WaferPresent	BOOLEAN	-	TRUE if wafer on chuck
1012	WaferID	STRING	-	Current wafer ID from OCR
1013	EMOStatus	BOOLEAN	-	Emergency Master Off state
1014	PumpStatus	STRING	-	Pump state: OFF/STARTING/RUNNING/FAULT

4. Data Variables

DVID	Name	Type	Unit	Description
2001	RecipeID	STRING	-	Active recipe name
2002	LotID	STRING	-	Current lot identifier
2003	WaferID_Processed	STRING	-	Wafer ID last processed
2004	EtchDepth	FLOAT	nm	Measured etch depth post-process
2005	EtchRate	FLOAT	nm/min	Calculated etch rate
2006	EtchTime_Actual	FLOAT	sec	Actual elapsed etch duration
2007	Uniformity	FLOAT	%	Within-wafer etch uniformity
2008	ProcessResult	STRING	-	PASS / FAIL / ABORTED

5. Event Report Guidance

The host should create one report for process start, one report for process end, and one alarm context report. Report IDs are host assigned and linked to the CEIDs in section 6 with S2F35/S2F37.

6. Collection Events

CEID	Name	Linked VIDs	Trigger
3001	ProcessStart	1001,1003,1004,2001,2002	Etch step begins
3002	ProcessEnd	2004,2005,2006,2008	Etch step ends
3003	ProcessAbort	1001,2008	Process aborted
3004	PlasmaIgnition	1005,1006	RF plasma ignited
3005	PlasmaExtinguish	1005	RF plasma off
3006	WaferLoaded	1011,1012	Wafer placed on chuck
3007	WaferUnloaded	1011,2008	Wafer removed from chuck
3008	AlarmSet	1001	Any alarm became active

7. Alarms

AlarmID	Name	Severity	Linked SVID	Description
4001	OverPressure	critical	1003	Chamber pressure exceeds limit
4002	PlasmaFailure	critical	1005,1006	Plasma failed to ignite
4003	EMO_Activated	critical	1013	Emergency stop activated
4004	PressureInstability	warning	1003	Pressure fluctuating >5%
4005	RF_HighReflection	warning	1006	Reflected power >10% of forward
4006	GasFlowDeviation	warning	1008	CF4 flow deviation >5% from setpoint

8. Remote Commands

RCMD	Description	Parameters
START_PROCESS	Start etch process	RECIPE_ID, WAFER_ID
ABORT_PROCESS	Abort active process	-
PAUSE_PROCESS	Pause at end of current step	-
RESUME_PROCESS	Resume paused process	-
ENTER_MAINTENANCE	Enter maintenance mode	-
EXIT_MAINTENANCE	Exit maintenance mode	-

9. Equipment State Machine

Nominal process flow is IDLE -> PUMPING -> PURGE -> PROCESSING -> COMPLETED -> IDLE. FAULT is reachable from PROCESSING and PURGE. Recovery requires operator acknowledgement and host-visible alarm clear.

From State	To State	Trigger
IDLE	PUMPING	START_PROCESS accepted
PUMPING	PURGE	Base pressure reached
PURGE	PROCESSING	Gas stabilization complete
PROCESSING	COMPLETED	EtchTime_Actual reached
COMPLETED	IDLE	WaferUnloaded
PROCESSING	FAULT	Critical alarm set
PURGE	FAULT	Pressure control failure
FAULT	IDLE	Alarm cleared and reset acknowledged

10. MES Tag Mapping Notes

Recommended MES tags include ChamberPressure, RFPower_Forward, RFPower_Reflected, RecipeID, LotID, WaferID_Processed, EtchDepth, EtchRate, Uniformity, ProcessResult, ProcessStart, ProcessEnd, AlarmSet, OverPressure, PlasmaFailure, and EMO_Activated.

11. Example SECS/GEM Transactions

Scenario	Primary Message	Expected Reply
Establish communications	S1F13	S1F14 COMMACK=0
Read selected variables	S1F3 <1003,1005,2001>	S1F4 with values
Enable collection events	S2F37 CEED=TRUE	S2F38 ACKC2=0
Start process	S2F41 RCMD=START_PROCESS	S2F42 HCACK=0
Alarm notification	S5F1 ALID=4001	S5F2 ACKC5=0

12. Revision History

Revision	Date	Description
A	2026-06-07	Initial demo interface specification